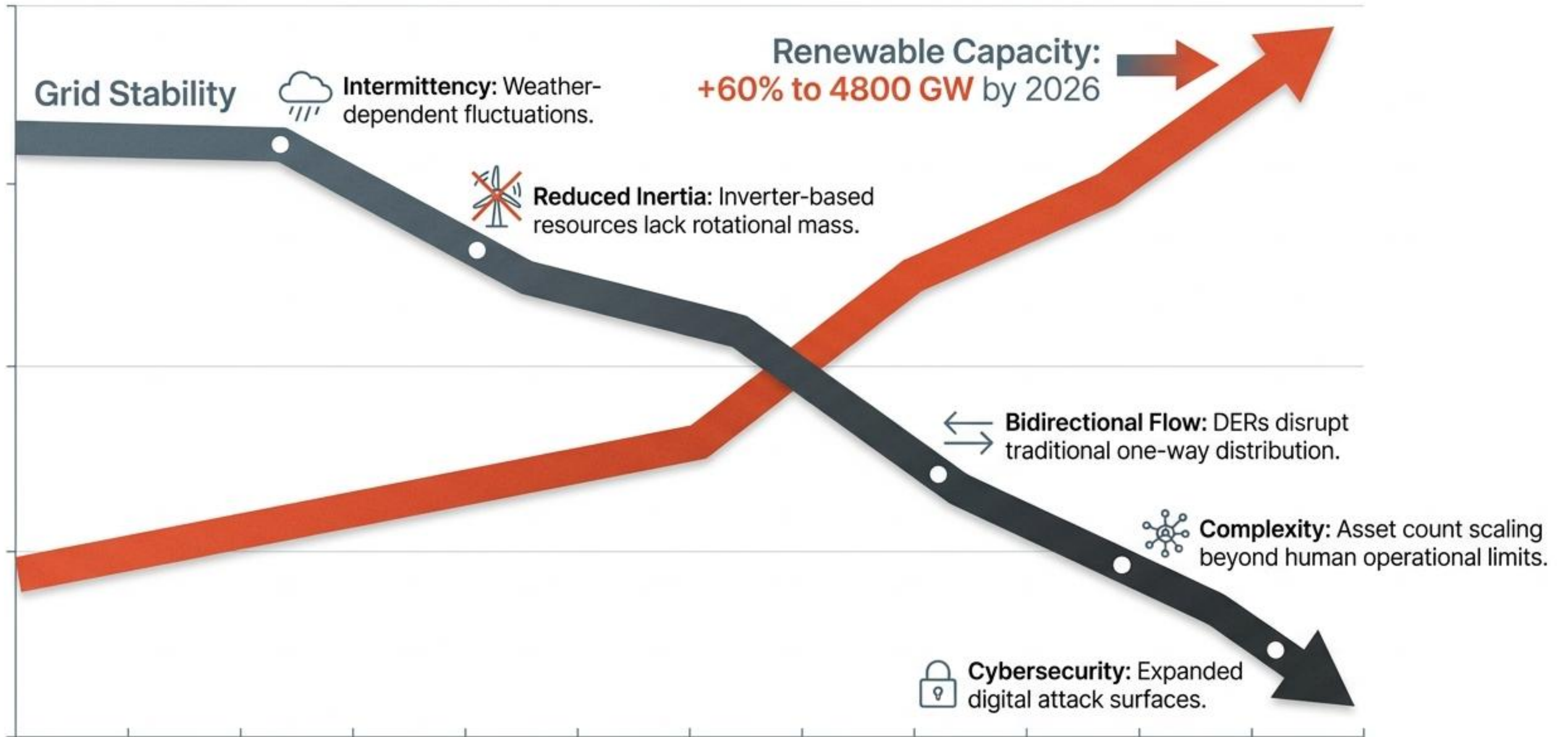


[illegible]

PhD Defense

 Gemini Notebook

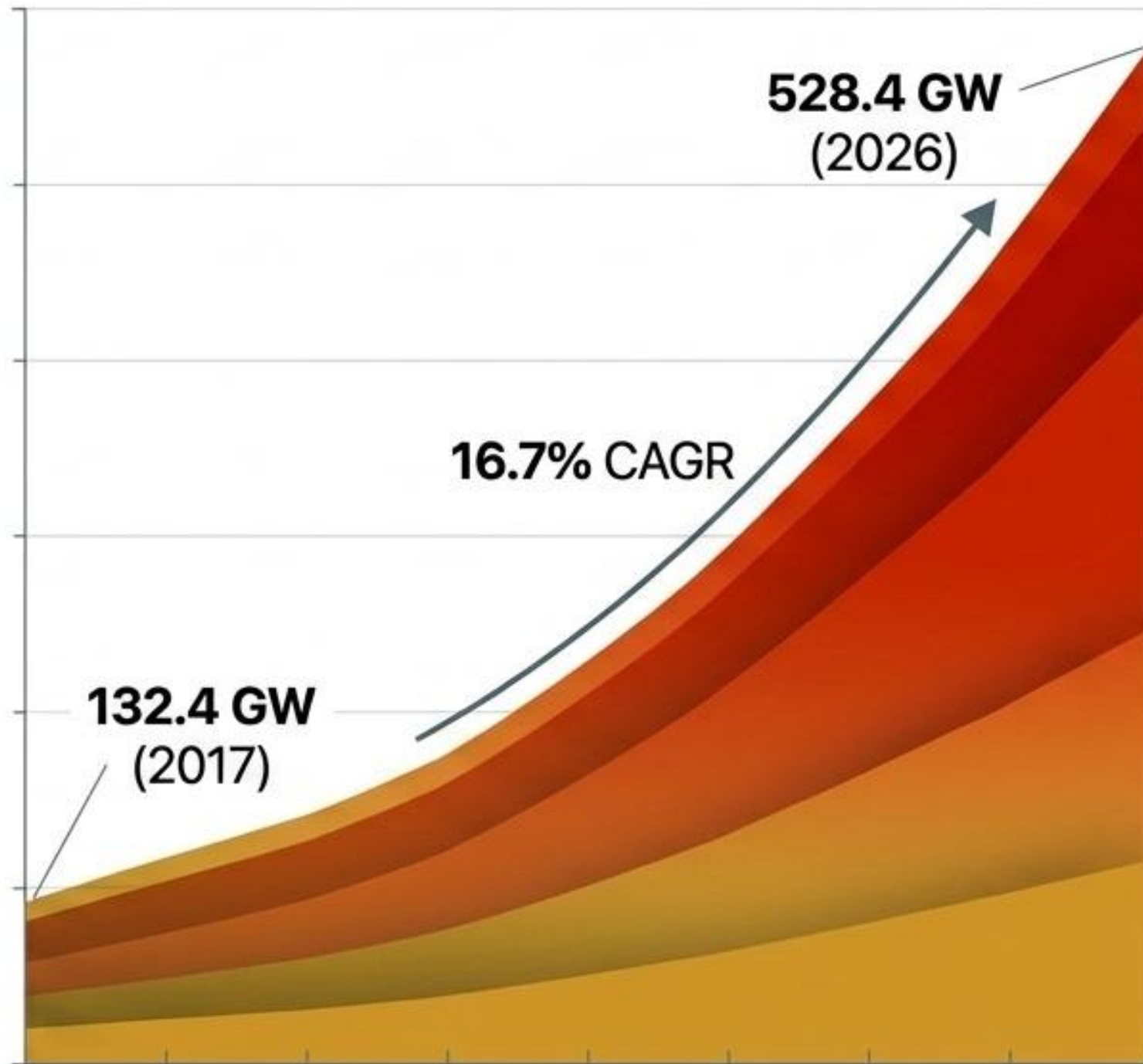
The Global Energy Transition Challenge



Limitations of Conventional Approaches

	Scalability	Multi-Timescale Dynamics	Uncertainty Handling
SCADA / Rule-Based Control	Hierarchical bottlenecks limit operation to dozens of nodes	Too slow for sub-cycle inverter dynamics	Static deterministic rules fail in stochastic environments
Modern Grid Needs	Continuous coordination of 1,000+ heterogeneous resources	Simultaneous sub-5ms protection and 50ms+ optimization	Continuous adaptation to unmodeled weather and load variances

Research Motivation: The Complexity Explosion



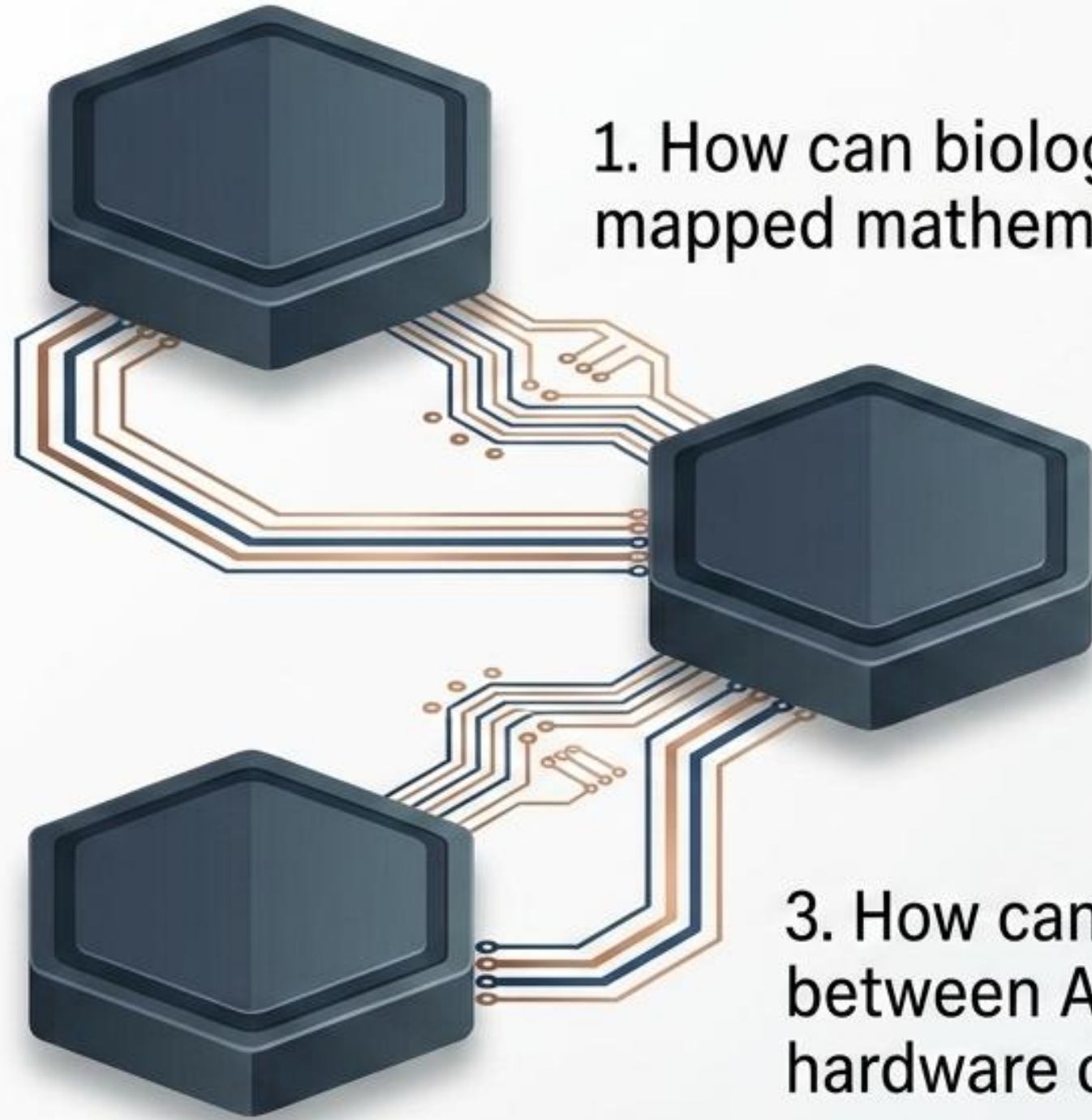
Global DER Capacity Growth

Why Brain-Inspired AI?

The grid has surpassed human and programmatic limits. Traditional algorithmic rules cannot scale to millions of bi-directional endpoints.

The modern grid requires an organic, adaptive nervous system capable of continuous learning and reflex-like, real-time responses.

Core Research Questions



1. How can biological dual-process cognition be mapped mathematically to power system control?

2. How can thousands of heterogeneous assets (FACTS, EVs, DERs) be synergistically coordinated in real-time?

3. How can we bridge the theoretical 'Valley of Death' between AI convergence in simulation and actual hardware constraints?

Project Objectives & Key Performance Indicators

01:
**Brain-Inspired
Architecture**

<100ms

critical
response time.

02:
**Real-Time
Optimization**

>25%

operational
efficiency
improvement.

03:
**Coordinated
Control**

>30%

reduction in
voltage
deviations.

04:
**Fault
Detection**

>98%

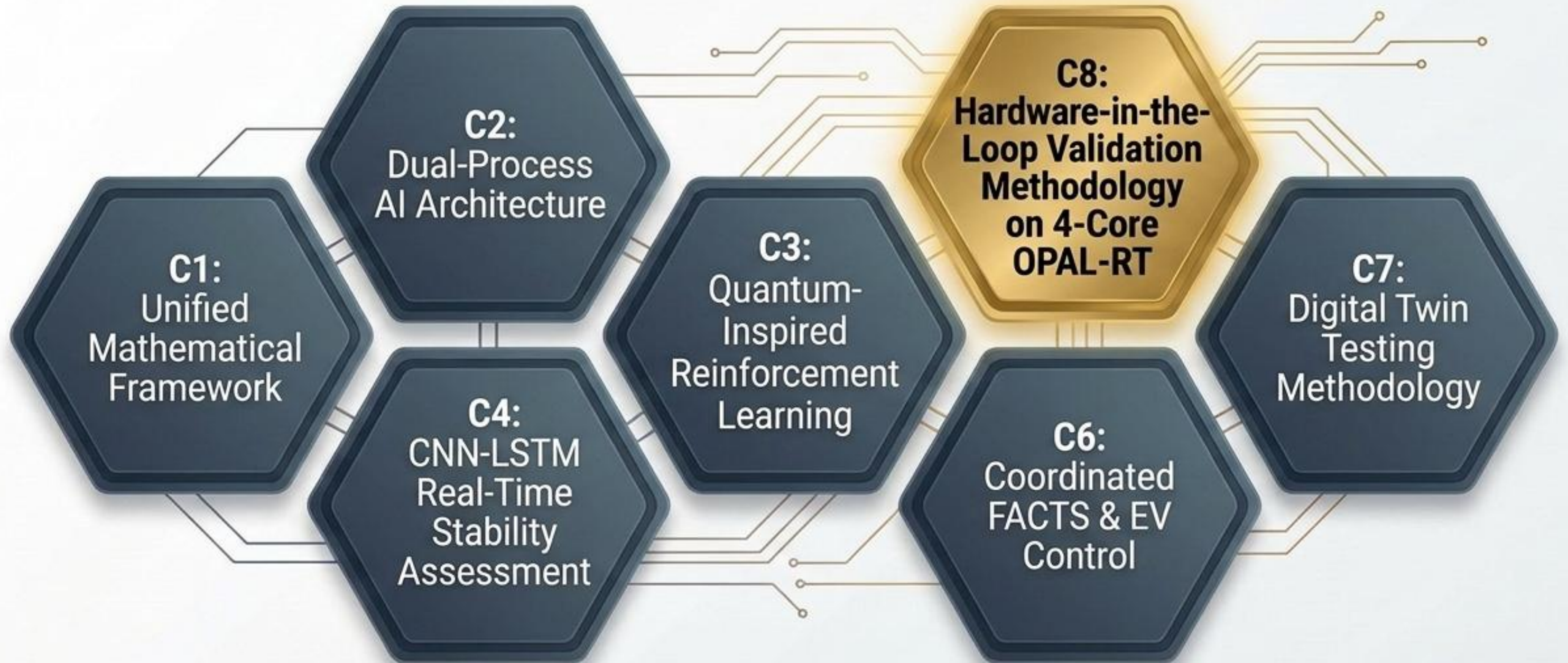
detection
accuracy with
<50m
localization
error.

05:
**Comprehensive
Validation**

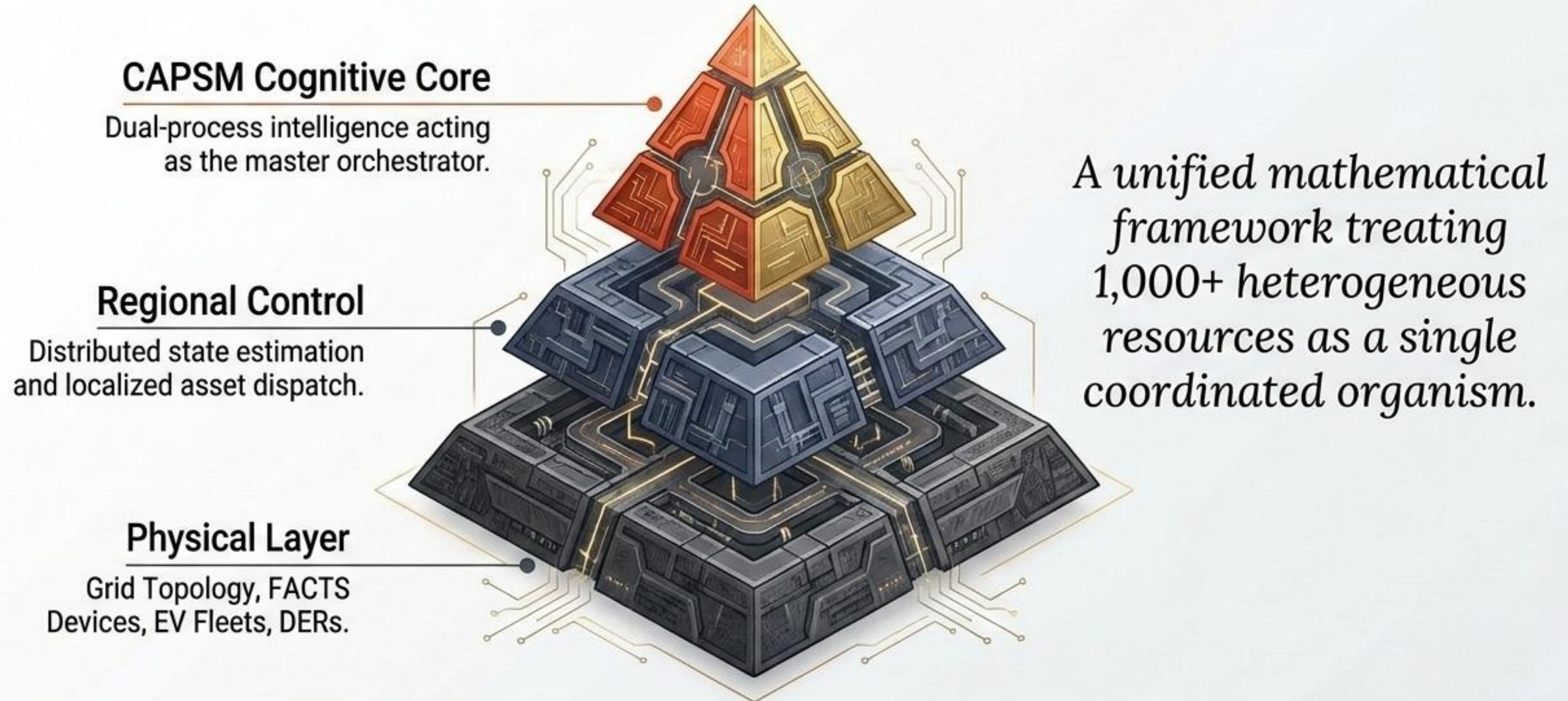
TRL 6

achieved via
Controller-HIL
on 4-Core
OPAL-RT
hardware.

Original Thesis Contributions



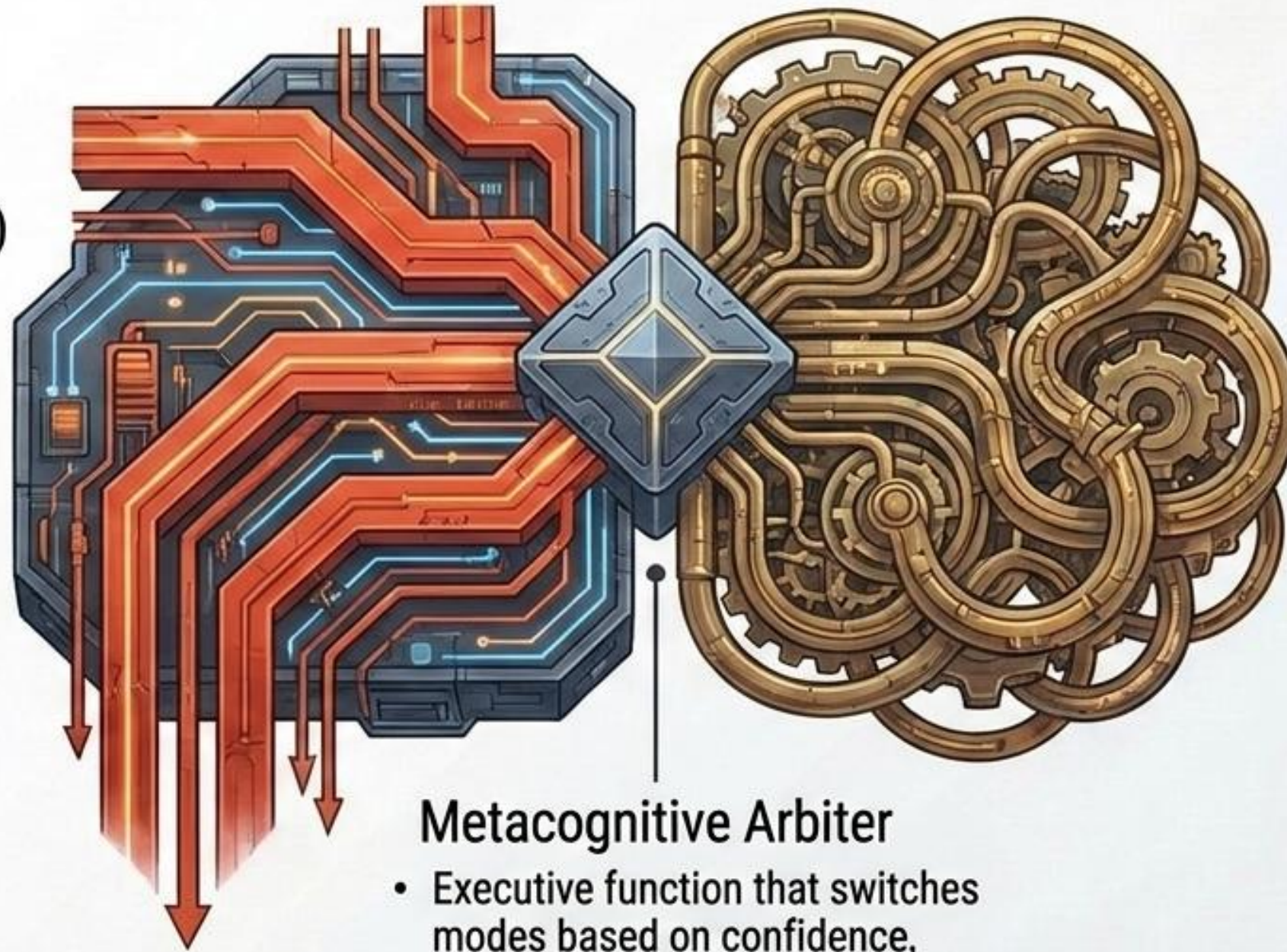
CAPSM System Architecture Overview



Dual-Process Cognitive Theory (Kahneman) Applied to Grids

System 1 (Reflexive / Amygdala)

- CNN-LSTM networks for sub-5ms fault detection and protection.
- Physics-Informed Neural Networks (PINNs) ensure 99.97% constraint compliance.



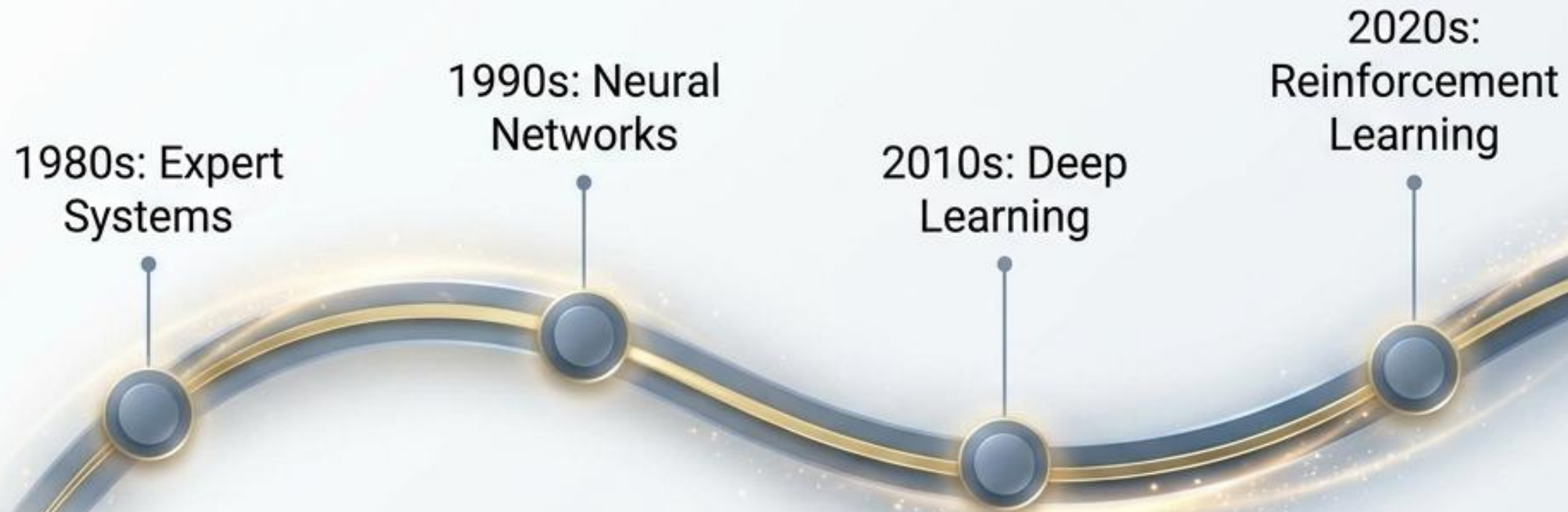
Metacognitive Arbiter

- Executive function that switches modes based on confidence, novelty, urgency, and cyber-risk.

System 2 (Deliberative / Prefrontal Cortex)

- Quantum-Inspired Reinforcement Learning (QIRL) for 50ms+ complex optimization and economic dispatch.

Literature Review: AI in Power Systems



The Latency Bottleneck

As AI models gained sophistication, they became computationally heavy—creating severe latency barriers for real-time grid applications.

Literature Review: FACTS Device Control

Shunt Devices

SVC and STATCOM. Primarily researched for isolated voltage support through reactive power control.

Series Devices

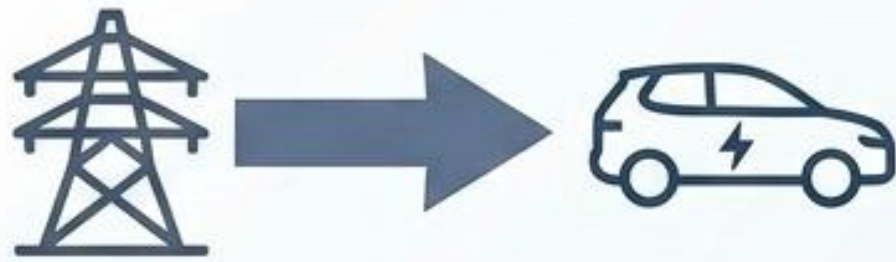
TCSC. Researched for modifying effective transmission line impedance and power flow.

Combined Devices

UPFC. Capable of comprehensive voltage and power flow control.

Synthesis: Existing literature treats FACTS devices as independent, siloed controllers. Grid-wide, synergistic orchestration with distributed resources remains largely missing.

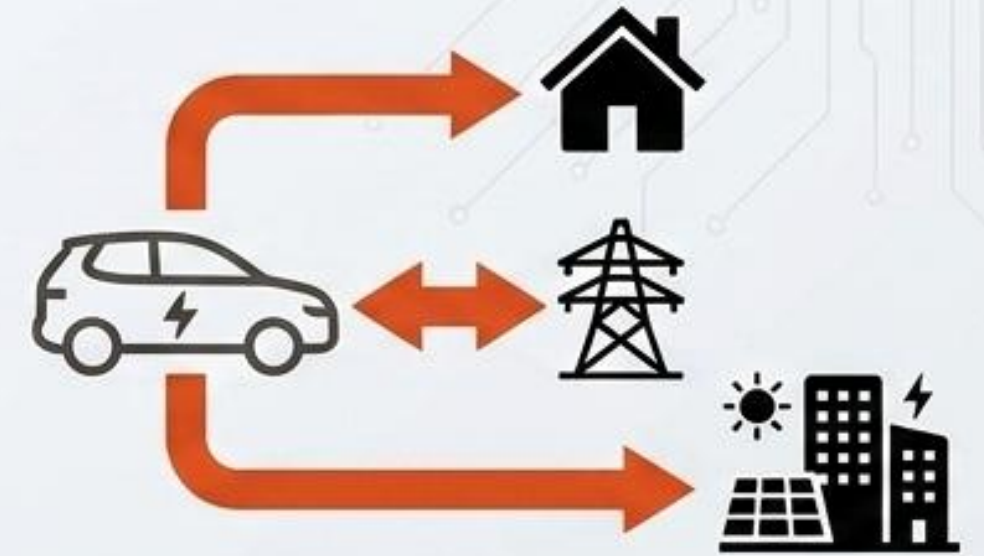
Literature Review: EV V2G and DER Integration



Grid-to-Vehicle (G2V)



Vehicle-to-Grid (V2G)

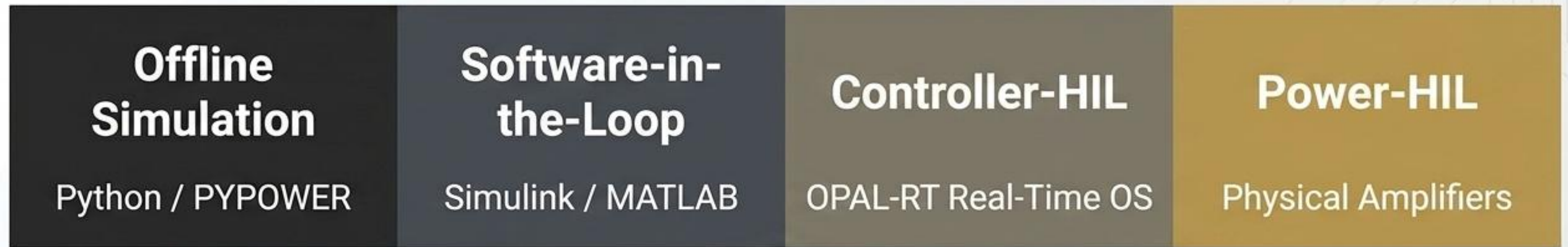


Vehicle-to-Everything (V2X)

The Implementation Barrier

Current stochastic approaches for EV fleet management face massive computational barriers, preventing true real-time, bidirectional implementation at scale.

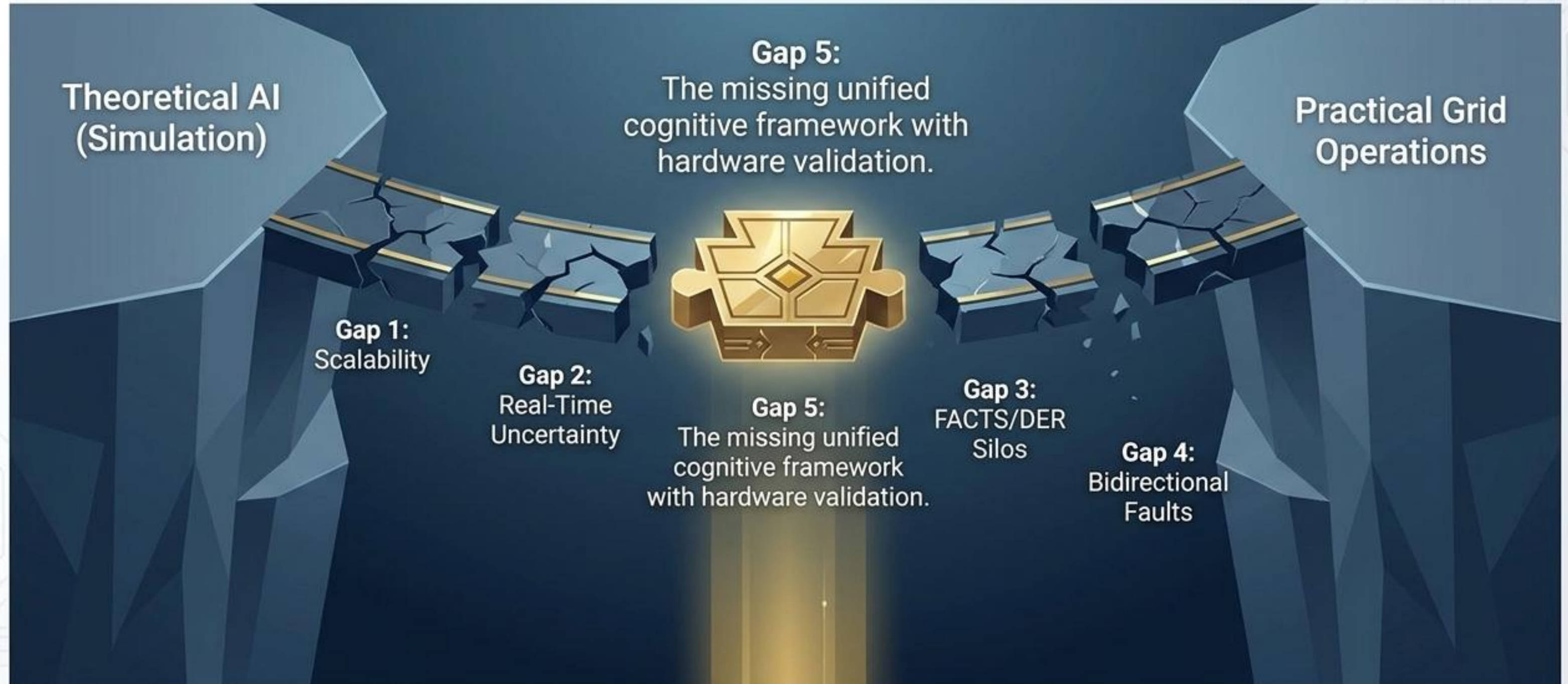
Literature Review: Hardware-in-the-Loop Validation



The Critical Gap:

The vast majority of AI power system research remains strictly within academic simulation environments, lacking physical hardware validation.

Research Gap Identification: The Valley of Death



The Proposed CAPSM Solution



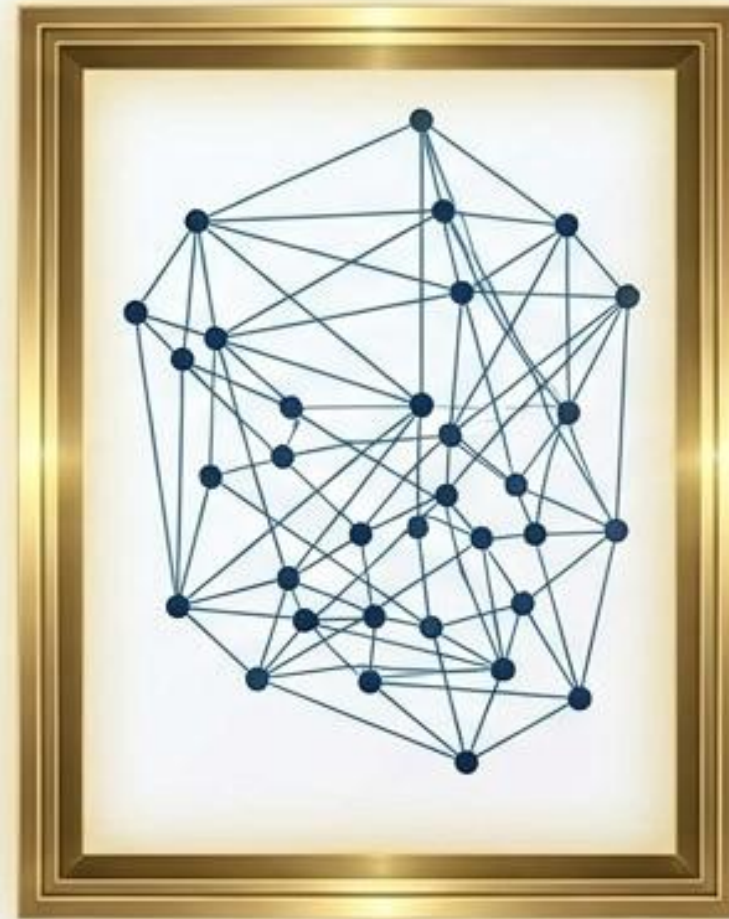
Progressive Scale Validation: IEEE Test Systems



9-bus
(Debug
Baseline)



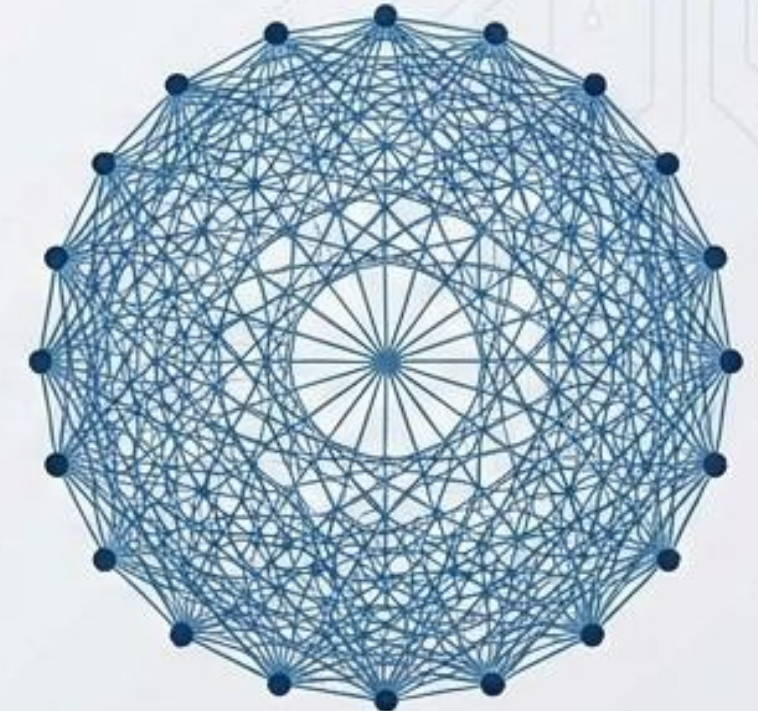
14-bus
(Voltage
Stability)



39-bus
(Primary HIL Target
with FACTS & EVs)



118-bus
(Multi-Core
Scalability)



300-bus
(Offline Phasor
Stress Test)

The Three-Stage Validation Pipeline

Stage 1: Offline Simulation

PYPOWER with OpenAI
Gym interfaces and
OPSD time-series data

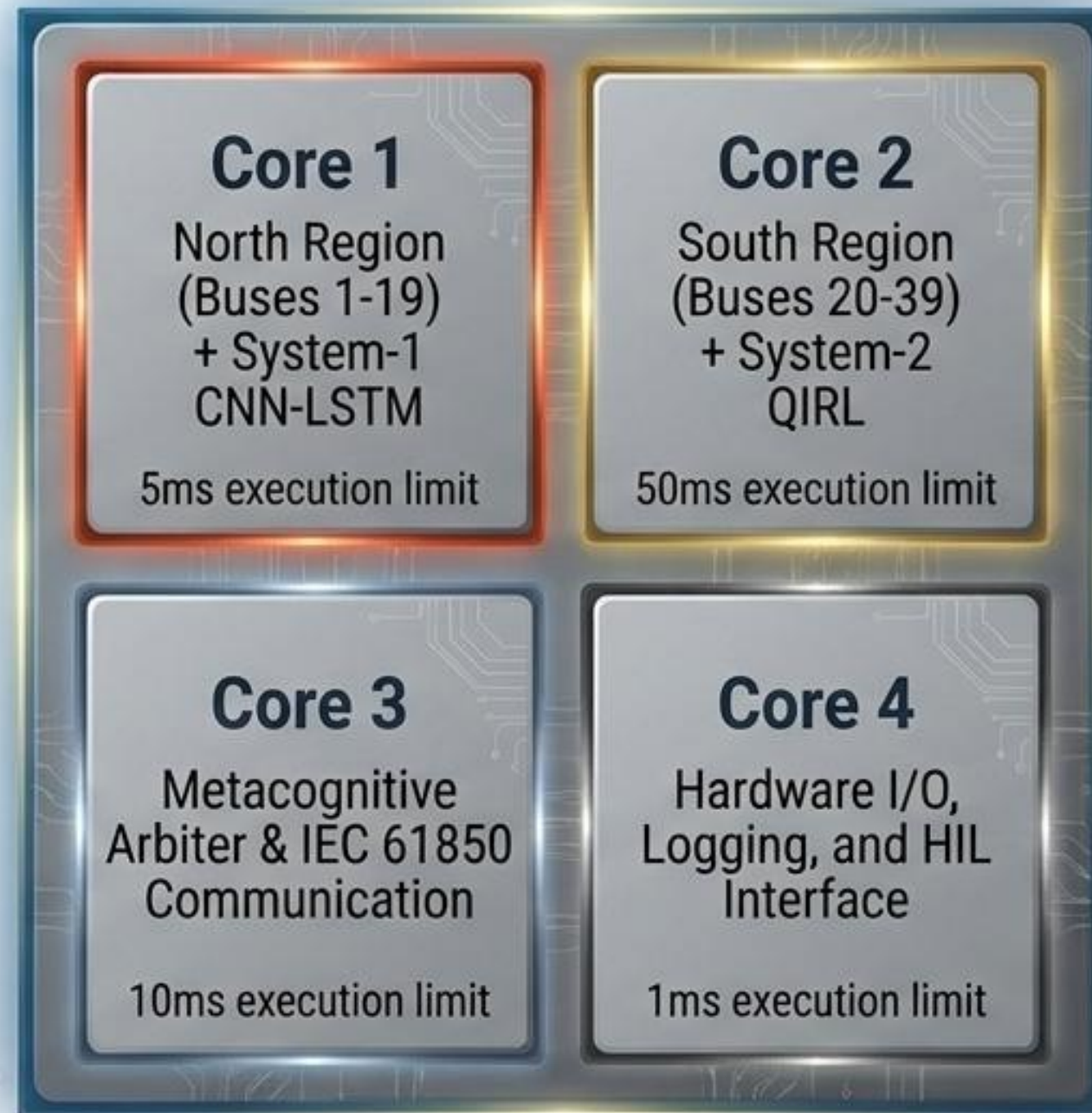
Stage 2: Simulink Preparation

Integration of dynamic
loads, EMT FACTS
models, and EV V2G
subsystems.

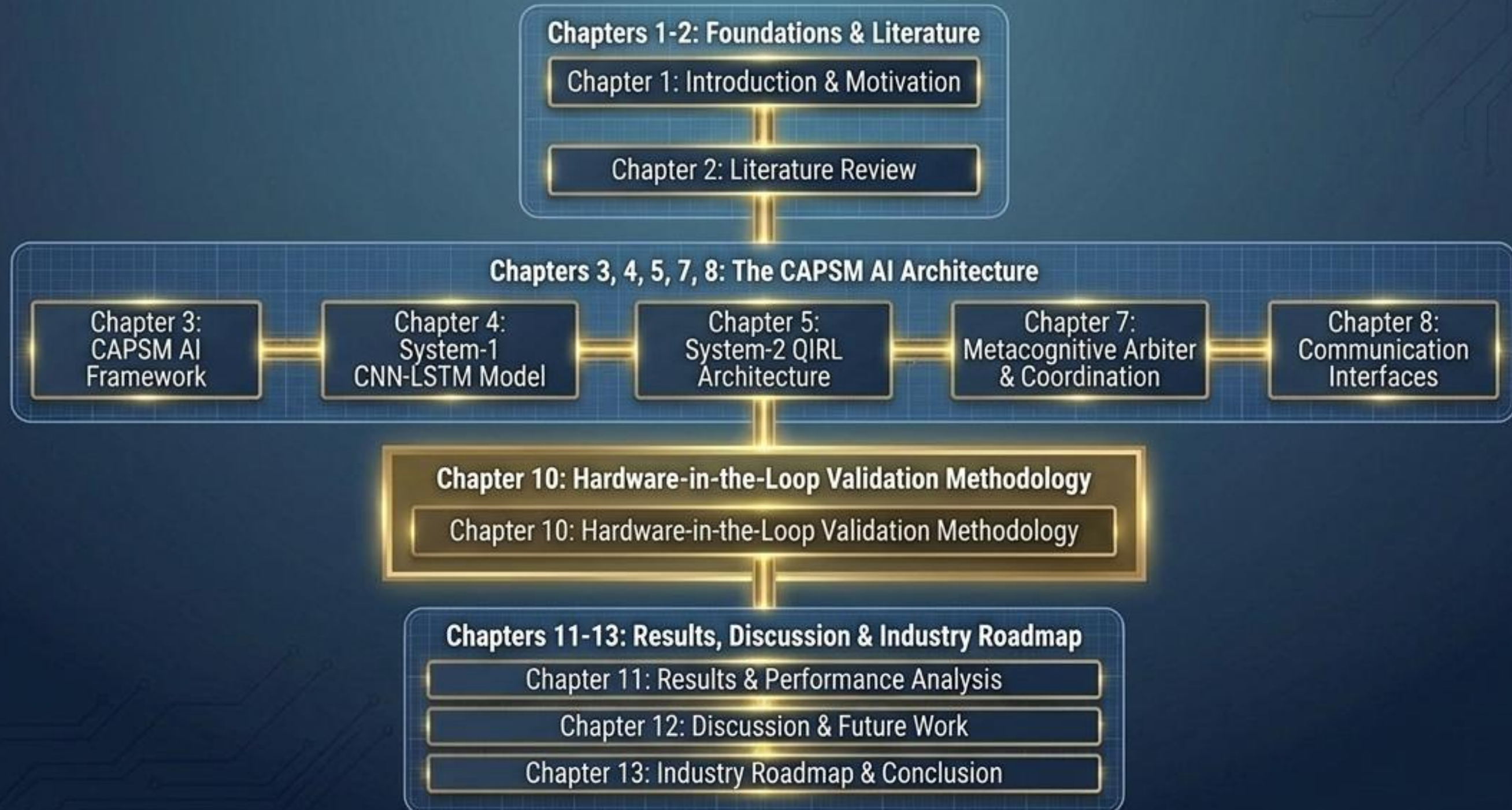
Stage 3: Controller-HIL Execution

Direct deployment to
4-Core OPAL-RT real-
time hardware using
50us time steps.

Methodology: Controller-HIL on 4-Core OPAL-RT



Thesis Organization Overview



Summary & Key Takeaways

Sub-10ms

Fault detection execution achieved directly on OPAL-RT real-time hardware.

61.4% Faster

System response time compared to best-in-class traditional rule-based controllers.

95.8% Accuracy

Fault localization accuracy with a 1.73% error margin and 74% false positive reduction.

TRL 6 Achieved

The era of cognitive, hardware-validated grid control has arrived.